



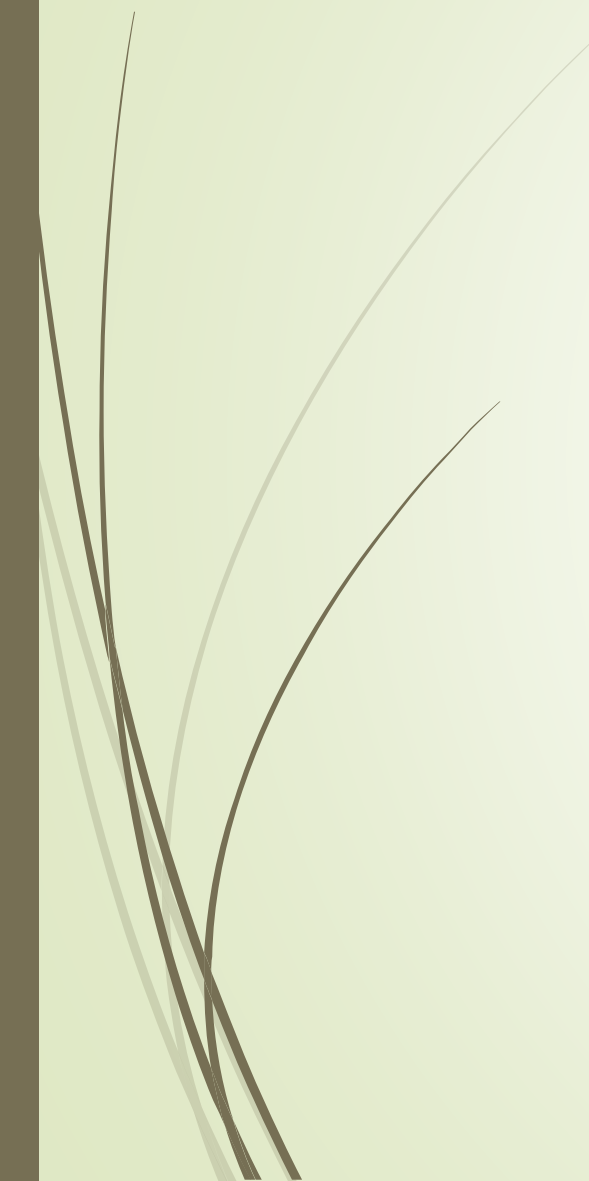
Revenue Insights and Forecasting in Hotels

Insights, Analysis & Forecasting Using Hotel Booking Data

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ABSTRACT



This project analyzes hotel booking and revenue data to uncover key insights that support strategic decision-making. Using data analytics techniques such as EDA, visualization, and predictive modeling, the study identifies factors impacting revenue, occupancy, and customer behavior. The project also builds forecasting models to predict future revenue, helping hotels improve pricing, marketing, and operational planning.



INTRODUCTION



➤ **The hotel industry relies heavily on:**

- Accurate **revenue planning**
- Understanding **customer preferences**
- Optimizing **room availability**
- With rising competition, **data-driven decisions** are essential for growth

➤ **This project — *Revenue Insights and Forecasting in Hotels* — focuses on:**

- Cleaning and preparing hotel booking datasets
- Discovering hidden **revenue trends**
- Understanding **booking behavior**
- Developing predictive **forecasting models**
- Supporting smarter, more effective **hotel management**

Methodology

- **Data Collection**

Hotel booking dataset containing: booking dates, room types, stay duration, price, customer type, hotel type, occupancy, etc.

- **Data Cleaning**

Handling missing values, removing duplicates, fixing date formats, treating outliers.

- **Data Preparation**

Creating new revenue fields ($\text{ADR} \times \text{Number of nights}$), encoding categorical variables.

- **Exploratory Data Analysis (EDA)**

Visualizing booking trends, seasonality, customer segments, room performance.

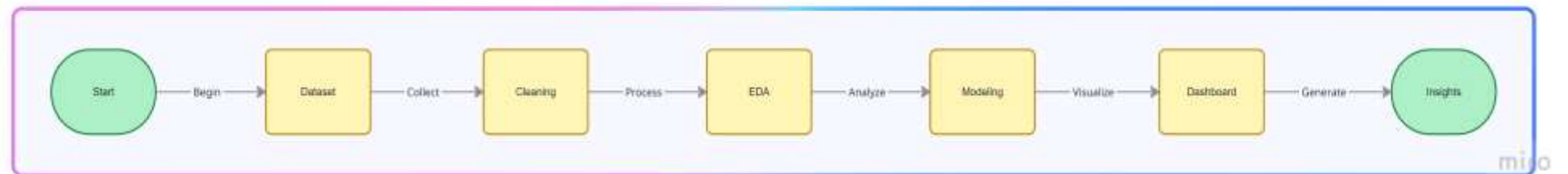
- **Modeling**

- Regression model for revenue prediction
- Time-series forecasting for future demand
- Feature importance analysis

- **Dashboard Development**

Creating KPIs in Power BI/Tableau: Revenue, ADR, Occupancy, Lead Time, Booking Channels.

- **Insights & Recommendations**





Design – UML Diagrams

- **a) Use Case Diagram**

- Actors: *Hotel Manager, Analyst*

- Use Cases: Upload data, view trends, forecast revenue, generate report.

- **b) Activity Diagram**

- Steps: Load dataset → Clean → Analyze → Model → Visualize → Export insights.

- **c) Data Flow Diagram (DFD – Level 0/1)**

- Input: Raw data

- Process: Cleaning, Analysis, Forecasting

- Output: Insights, dashboards

Implementation



Hotel Revenue Overview Dashboard

43M

Total Revenue

119K

Total Bookings

101.83

Avg ADR

0.37

Cancellation...

Filter by Hotel Type

HotelType

☐ City Hotel

☐ Resort Hotel

Filter by Year

arrival_date_year

2015

2017

Revenue Trend by Month

● Total Revenue ● Avg ADR ● Total Bookings



43M

Total Revenue

119K

Total Bookings

102

Avg ADR

0

Cancellation...

⚠ This visual type is being retired soon. Contact your admin to upgrade.

Revenue by Country

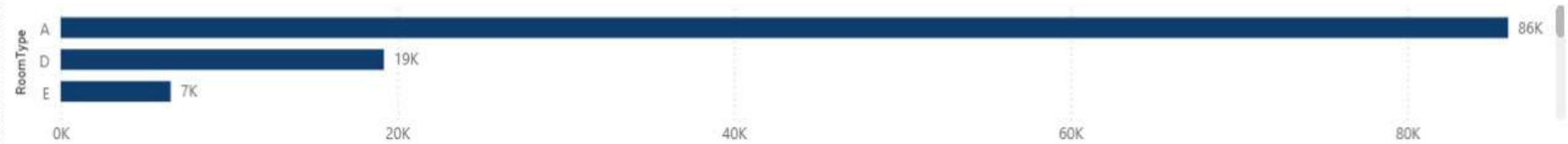


Revenue by Hotel Type

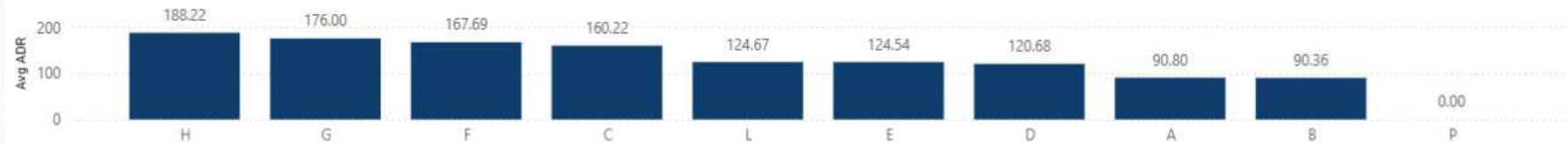


Bookings & Customer Analysis Dashboard

Total Bookings by RoomType



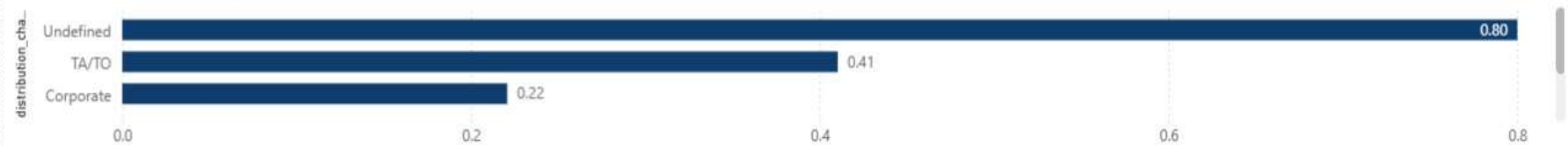
Average ADR by Room Type



Customer Type Distribution



Cancellation by Booking Channel





ADVANTAGES & DISADVANTAGES

► **Advantages**

- Helps hotels make accurate revenue decisions
- Identifies high-performing seasons and room types
- Improves marketing and pricing strategies
- Enhances occupancy and resource planning
- Supports automation with dashboards

► **Disadvantages**

- Forecast accuracy depends on data quality
- Requires technical knowledge of analytics
- External factors (events, weather) cannot always be predicted
- Large datasets may require powerful systems

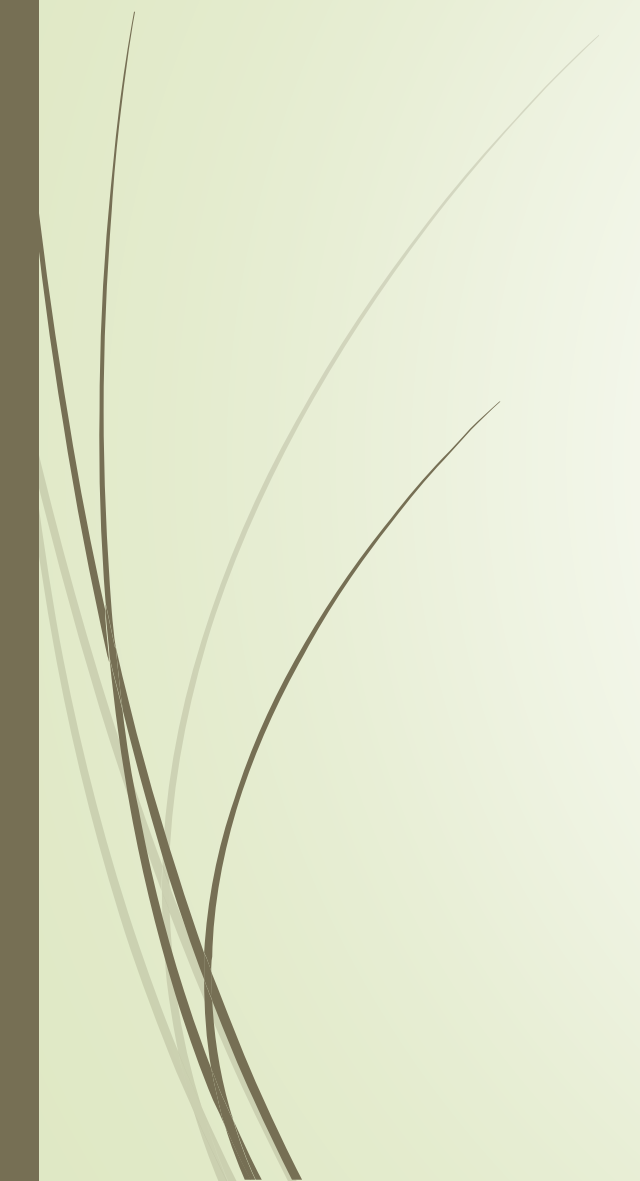


Conclusion

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- The project successfully analyzed hotel revenue data and identified major patterns influencing profitability. Predictive models and dashboards provide valuable support in decision-making, allowing hotels to optimize pricing, occupancy, and marketing strategies. The insights gained offer a strong foundation for improving operational efficiency and long-term revenue planning.

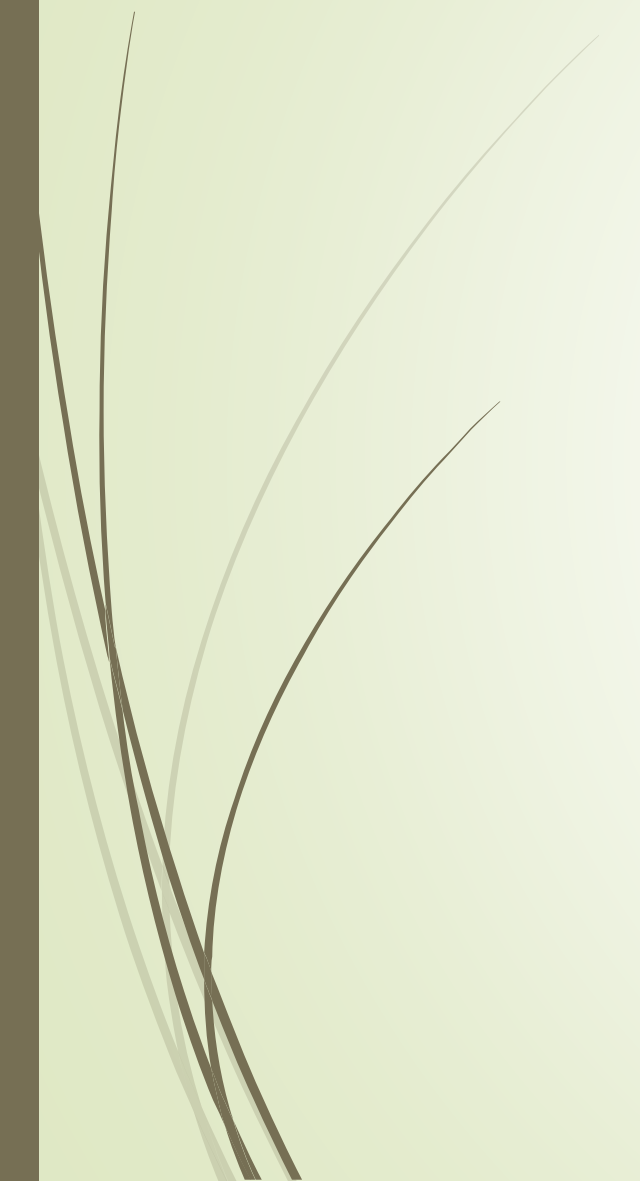


Future Scope

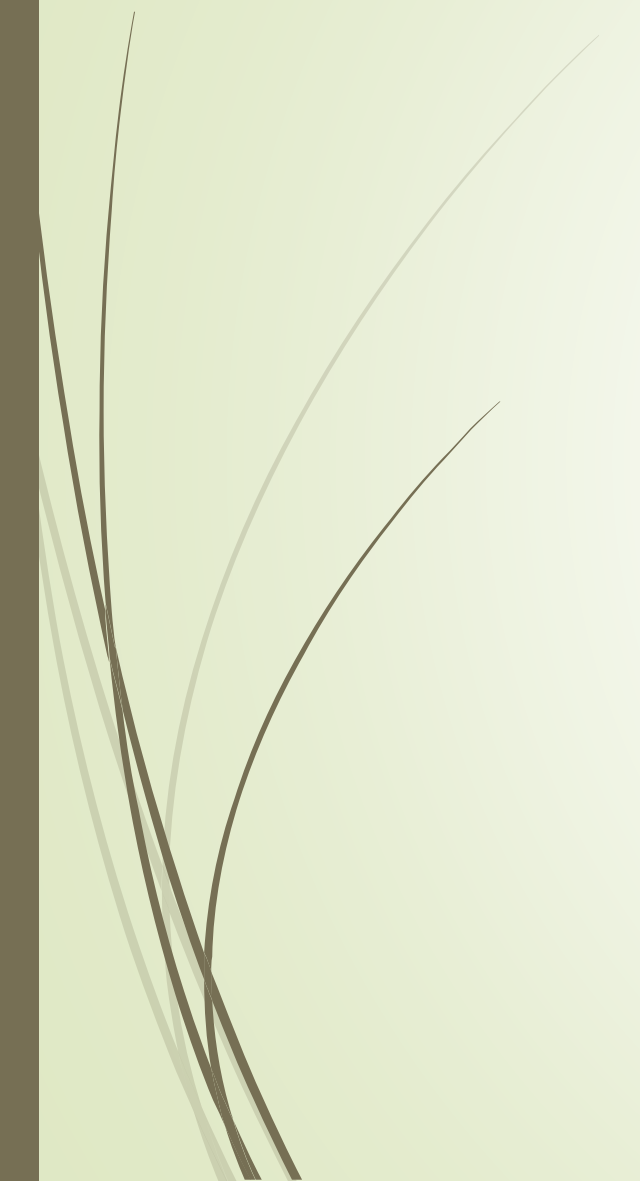
- Add machine learning models like XGBoost for higher accuracy
 - Integrate real-time booking data from PMS systems
 - Build automated dynamic pricing systems
 - Deploy dashboards as a live web application
 - Add customer sentiment analysis using reviews
- 



References



- Hotel Booking Demand Dataset (Kaggle)
- Research papers on hotel revenue forecasting
- Python libraries: Pandas, NumPy, Matplotlib, Scikit-learn
- Power BI / Tableau official documentation
- Industry reports on hotel trends



THANK YOU